

$$A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$$

$$\text{najdi } A^* A = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 6 & -3 & -3 \\ 0 & -3 & 6 & -3 \\ 0 & -3 & -3 & 6 \end{pmatrix}$$

nalezni vlastní čísla a vlastní vektory

$$\lambda_1 = 9, M_9 = \text{span} \left\{ \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\lambda_2 = 3, M_3 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$\lambda_3 = 0, M_0 = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

nalezni singulární hodnoty

$$\sigma_1 = \sqrt{\lambda_1} = 3$$

$$\sigma_2 = \sqrt{\lambda_2} = \sqrt{3}$$

$$D = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}$$

V sestavím normalizovaným vl. vektory

$$V = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$$

U matricu ponovo vratim

$$u_i = \frac{Av_i}{\|Av_i\|}$$

$$u_1 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & -2 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{3\sqrt{2}} \begin{pmatrix} -3 \\ 0 \\ 3 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

$$u_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ -3 \\ 3 \\ 3 \end{pmatrix} =$$

a dopunim na ON bazi

$$u_3 \in \left(\text{span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -3 \\ 3 \\ 3 \end{pmatrix} \right\} \right)^\perp = \text{Ker} \begin{pmatrix} -1 & 0 & 1 \\ 0 & -3 & 3 \end{pmatrix} =$$

$$= \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$u_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$A = \left(\frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \middle| \frac{1}{\sqrt{6}} \begin{pmatrix} 0 \\ -3 \\ 3 \\ 3 \end{pmatrix} \middle| \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} (0 \ -1 \ 0 \ 1) \\ \frac{1}{\sqrt{2}} (0 \ -1 \ 1 \ 0) \\ \frac{1}{\sqrt{3}} (1 \ 1 \ 1) \end{pmatrix}$$

Prv A^T transformiraj

$$A^T = \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \middle| \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix} \middle| \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \middle| \frac{1}{\sqrt{3}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right) \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} (-1 \ 0 \ 1) \\ \frac{1}{\sqrt{2}} (0 \ -3 \ 3) \\ \frac{1}{\sqrt{3}} (1 \ 1 \ 1) \end{pmatrix}$$

